



Coal and Petroleum

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Coal & Petroleum

Comprehensive Guide for Olympiad
Preparation

Introduction to Fuels

- Definition:** A combustible substance which produces a large amount of heat and light on burning.
- Examples:** Coal, LPG, Petrol, Kerosene, Wood

Classification of Fuels

- **Basis:** Physical state
- **Types:**
 - Solid Fuels
 - Liquid Fuels
 - Gaseous Fuels

Solid Fuels

- Definition:** Combustible substances solid at room temperature.
- Examples:** Coal, Coke, Wood, Charcoal, Animal-dung cakes, Bagasse
- Usage:** Mainly rural; includes firewood and agricultural wastes

Liquid Fuels

- **Definition:** Volatile liquids that produce combustible vapor
- **Examples:** Petrol, Diesel, Kerosene, Alcohol
- Produce high energy when burned.

Gaseous Fuels

- **Definition:** Combustible gases or mixtures
- **Examples:** Natural gas, Biogas, LPG, CNG, Producer gas, Water gas, Hydrogen

Characteristics of an Ideal Fuel

- High calorific value
- Moderate ignition temperature
- No harmful residue
- Low cost & easily available
- Easy to transport and store
- Combusts efficiently

Uses of Fuels

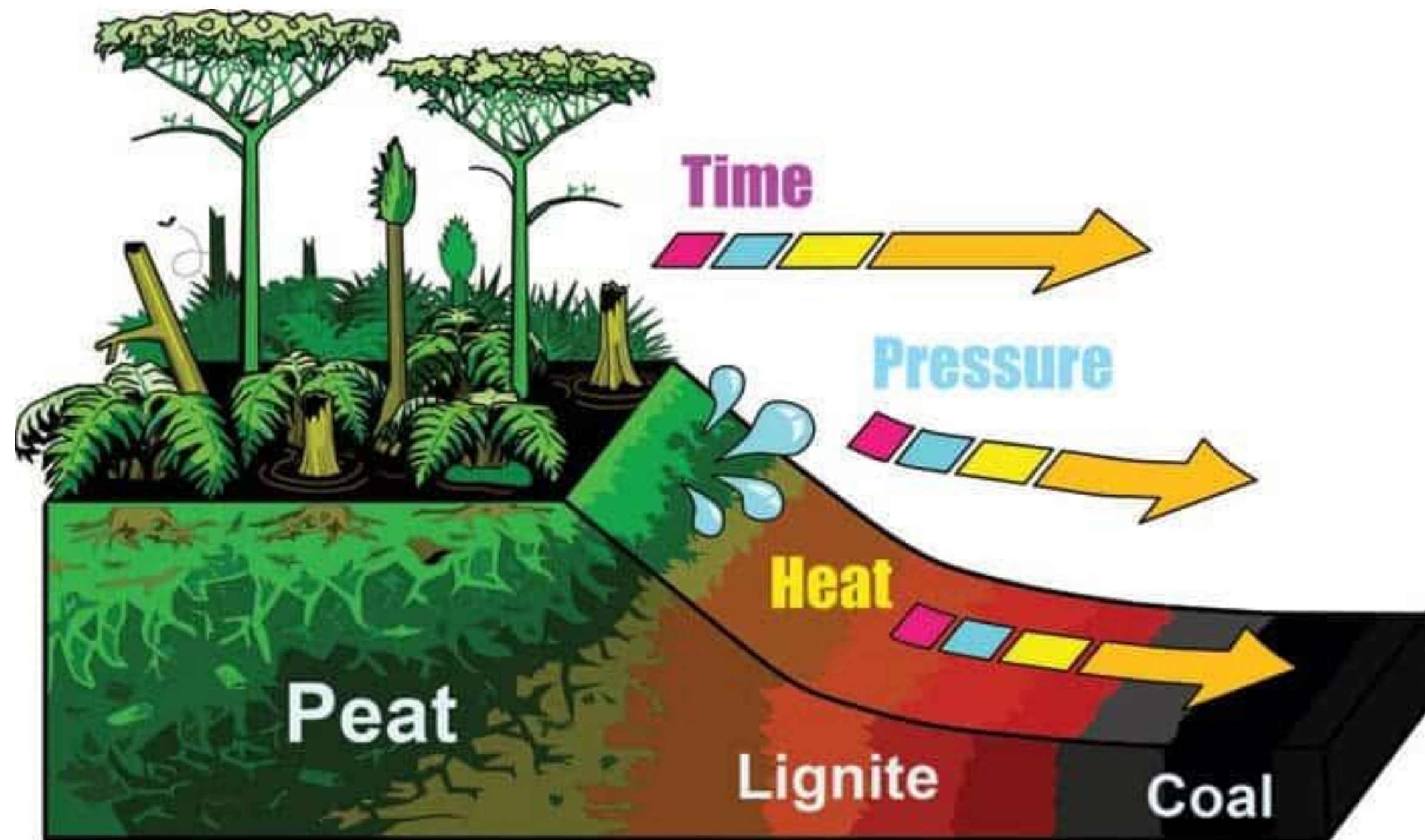
- **Cooking & Heating:** Wood, dry cattle dung, charcoal, kerosene, LPG.
- **Transportation:** Petrol, diesel, CNG used in cars, buses, trucks, scooters, airplanes.
- **Electricity Generation:** Used in thermal power stations.
- **Industry:** Coal, natural gas, and diesel generate steam.
- **Space Launching:** Special fuels called **propellants** are used.

Coal – Introduction

- **Definition:** Black or brown mineral from remains of buried plants
- **Composition:** Mainly carbon, hydrogen, oxygen; small sulfur
- **Mining types:** Open-cast(near) & Underground(far)

Formation of Coal

- From plants buried millions of years ago in swampy regions
- Heat & pressure → Coal

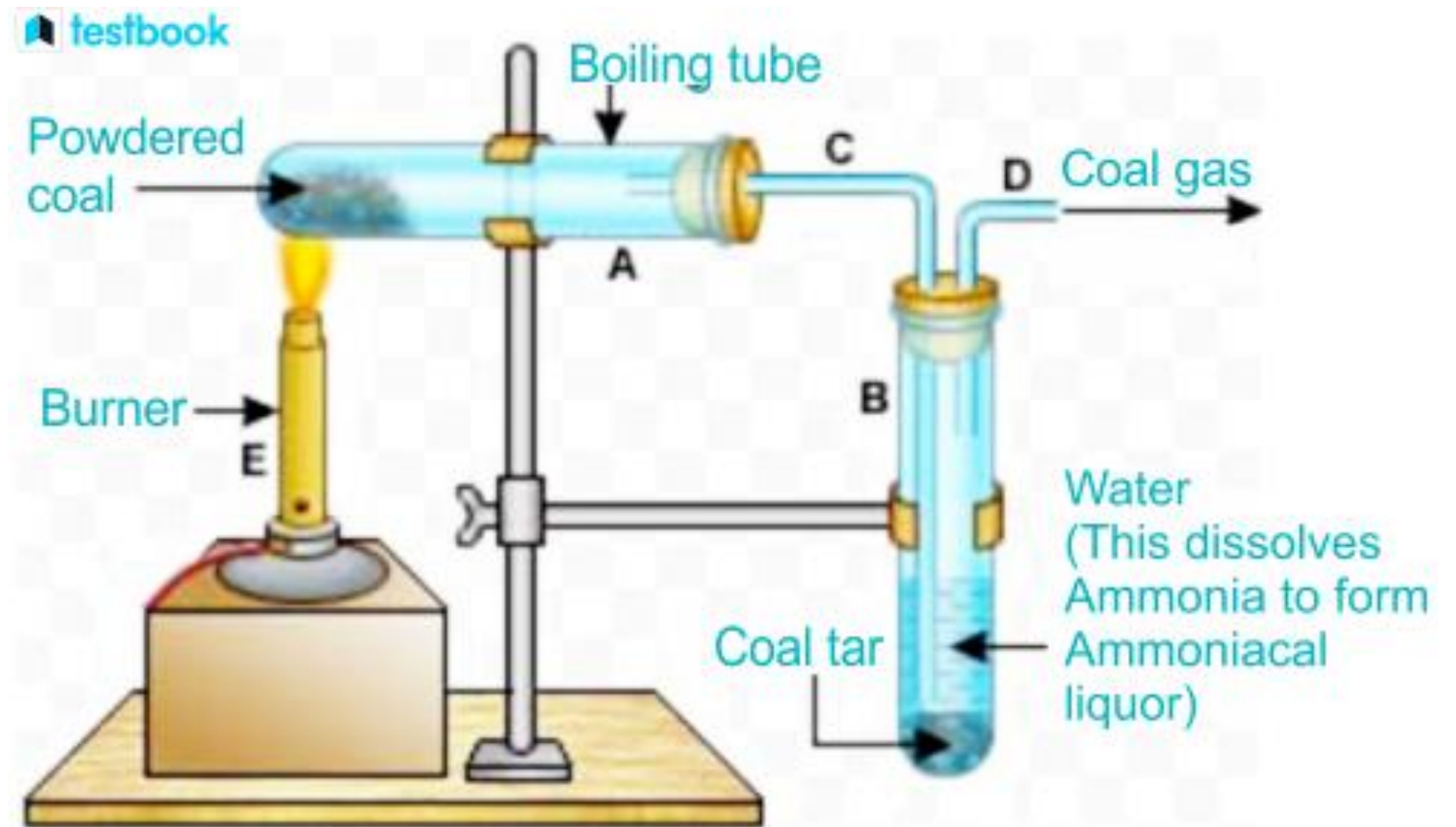


Types of Coal

- **Peat:** Low carbon, high moisture, 27%
- **Lignite:** 28–30% carbon, brownish-black
- **Bituminous:** 78–86% carbon, most used
- **Anthracite:** 94–98% carbon, superior quality

Destructive Distillation of Coal(pyrolysis)

- High temp heating without air (around 1270 K)
- Products:**
 - Coke
 - Coal tar
 - Ammoniacal liquor
 - Coal gas



Destructive Distillation

Uses of Coal Products

- **Coke:**

- Used in domestic fuel
- Used in steel manufacturing as reducing agent

- **Coal Tar:**

- Thick black liquid with various chemicals
- Used in roads, explosives, synthetic dyes, plastics

- **Ammoniacal Liquor:**

- Contains ammonia
- Used in fertilizers

Coal Gas:

- Hydrogen + methane + carbon monoxide
- Used for lighting (historically), heating

Petroleum – Introduction

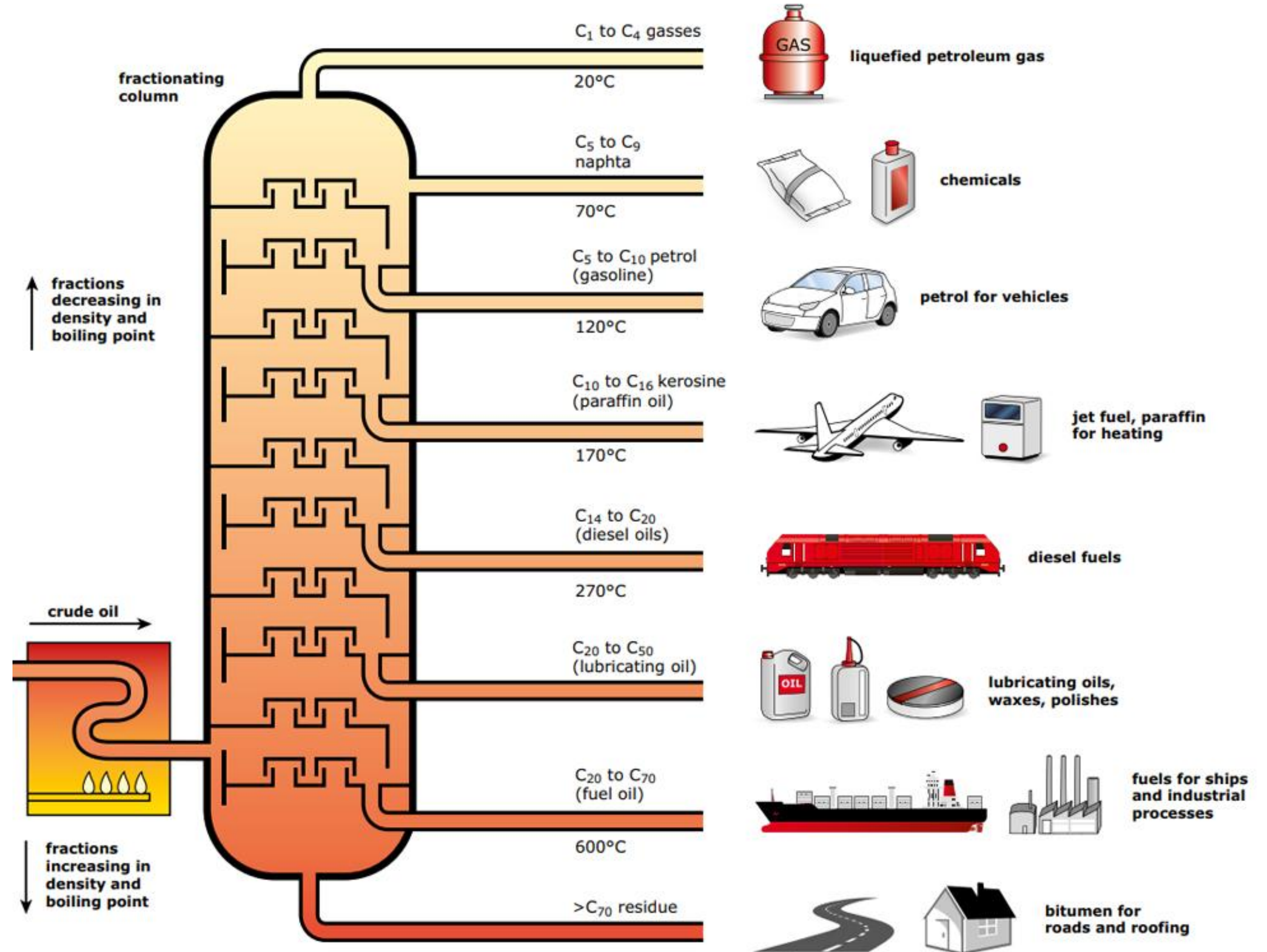
- Naturally occurring oil with hydrocarbons
- Found in sedimentary rock under pressure
- **Crude oil = Petroleum**

Origin & Extraction

- Formed from remains of sea organisms.
- Buried under mud, clay, sand; converted to oil by heat/pressure.
- Oil is trapped in porous rocks under non-porous rock.
- **Major Indian oil fields:**
 - Digboi (Assam)
 - Ankleshwar & Kalol (Gujarat)
 - Offshore areas near Mumbai

Refining of Petroleum

- **Definition:**
Separation by fractional distillation based on boiling points



Products of Petroleum Refining

- **Asphalt:** Road surfacing
- **Paraffin Wax:** Candles, vaseline, grease, polish
- **Lubricating Oil:** Machines
- **Furnace Oil:** Boilers
- **Diesel Oil:** Cars, trucks, buses
- **Kerosene:** Lamps, heaters, jets
- **Petrol:** Cars, scooters, two-wheelers
- **Residual Oil:** Used in insulation, paints

Petroleum Gas & LPG

- **Petroleum Gas:** Butane + Propane
 - Stored under pressure in LPG cylinders
- **LPG:** Liquefied Petroleum Gas
 - Odorless but ethyl mercaptan added for smell ($\text{C}_2\text{H}_6\text{S}$)

Petrochemicals

- Derived from petroleum and natural gas
- Used to make:
 - Plastics
 - Fertilizers
 - Insecticides
 - Synthetic fibers
 - Detergents
 - Medicines
- Called **Petrochemicals Industry**

